

EDUCATION

- **Hong Kong University of Science and Technology (HKUST)** Hong Kong, China
Ph.D. student in Computer Science and Engineering. Supervisor: Prof. Dan Xu Feb. 2022 – Present
- **University of Science and Technology of China (USTC)** Hefei, China
Bachelor of Science in Automation Sep. 2015 – June 2019

EXPERIENCE

- **Tencent HY Team** Shenzhen, China
Research Intern May 2025 – Present
 - **3D Vision-Language Models:** As first author, led the end-to-end development of N3D-VLM for native 3D grounding and spatial reasoning. Built a scalable pipeline to lift 2D annotations into 3D grounding and spatial-QA supervision, and designed the unified model and evaluation. Also contributed to DepthVLM and Stream3D-VLM for metric depth estimation and online 3D understanding.
 - **Computer-Use Agents:** Developed OSWorkerBench for long-horizon, cross-application GUI agents. Designed and validated benchmark tasks in interactive environments with executable evaluators, using human-in-the-loop verification with agent rollouts. Corresponding demonstrations were designed for the tasks. Also constructed long-horizon training data and contributed to DemoCUA harness design.
- **Shanghai AI Lab** Shanghai, China
Research Intern Feb. 2022 – Feb. 2023
 - **Datasets & Benchmarks:** Co-developed OmniObject3D (**CVPR 2023, Award Candidate: 12/9,155**) and RenderMe-360 (**NeurIPS 2023**), two large-scale 3D/4D datasets and benchmarks. Contributed to video data collection, data processing pipelines, benchmark construction, and evaluation.
- **SenseTime Inc.** Beijing, China
Research Intern $\Rightarrow$ Computer Vision Researcher Sep. 2019 – Feb. 2022
 - **Text/Audio-Driven Talking Heads:** Developed and deployed text- and audio-driven 2D, 3D, and cartoon head generation algorithms across multiple products, generating more than RMB 10 million in revenue for the company. The work resulted in three patents and a first-author book chapter.

SELECTED PUBLICATIONS

Vision-Language Models & GUI Agents

- **Yuxin Wang**, Lei Ke, Boqiang Zhang, Tianyuan Qu, Hanxun Yu, Zhenpeng Huang, Meng Yu, Dan Xu, Dong Yu. “N3D-VLM: Native 3D Grounding Enables Accurate Spatial Reasoning in Vision-Language Models.” **arXiv 2025**.
- Tencent HY Frontier Team (including **Yuxin Wang** as a Core Contributor). “UI-Mate: Advancing Open-Weight Foundation GUI Agents with In-Context Demonstrations.” **arXiv 2026**.
- Hanxun Yu*, Xuan Qu*, **Yuxin Wang**, Jianke Zhu, Lei Ke. “Unlocking Dense Metric Depth Estimation in VLMs.” **arXiv 2026**.
- Hanxun Yu*, Xuan Qu*, Lei Ke, Boqiang Zhang, **Yuxin Wang**, Jianke Zhu, Dong Yu. “Stream3D-VLM: Online 3D Spatial Understanding with Incremental Geometry Priors.” **ECCV 2026**.

3D Generation, Reconstruction & Editing

- **Yuxin Wang**, Qianyi Wu, and Dan Xu. “F3D-Gaus: Feed-forward 3D-aware Generation on ImageNet with Cycle-Aggregative Gaussian Splatting.” **arXiv 2025**.
- **Yuxin Wang**, Qianyi Wu, Guofeng Zhang, and Dan Xu. “GScream: Learning 3D Geometry and Feature Consistent Gaussian Splatting for Object Removal.” **ECCV 2024**.
- **Yuxin Wang**, Wayne Wu, and Dan Xu. “UDC-NeRF: Learning Unified Decompositional and Compositional NeRF for Editable Novel View Synthesis.” **ICCV 2023**.
- Zhenxing Mi, **Yuxin Wang**, Dan Xu. “One4D: Unified 4D Generation and Reconstruction via Decoupled LoRA Control.” **ECCV 2026**.
- Shaocong Dong, Lihe Ding, Xiao Chen, Yaokun Li, **Yuxin Wang**, Yucheng Wang, Qi Wang, Jaehyeok Kim, Chenjian Gao, Zhanpeng Huang, Zibin Wang, Tianfan Xue, Dan Xu. “From One to More: Contextual Part Latents for 3D Generation.” **ICCV 2025**.

Datasets & Benchmarks

- Tong Wu, Jiarui Zhang, Xiao Fu, **Yuxin Wang**, Jiawei Ren, Liang Pan, Wayne Wu, Lei Yang, Jiaqi Wang, Chen Qian, Dahua Lin, and Ziwei Liu. “OmniObject3D: Large-Vocabulary 3D Object Dataset for Realistic Perception, Reconstruction and Generation.” **CVPR 2023 (Award Candidate: 12/9155)**.
- Dongwei Pan, Long Zhuo*, Jingtian Piao*, Huiwen Luo*, Wei Cheng*, **Yuxin Wang***, Siming Fan, Shengqi Liu, Lei Yang, Bo Dai, Ziwei Liu, Chen Change Loy, Chen Qian, Wayne Wu, Dahua Lin, and Kwan-Yee Lin. “RenderMe-360: A Large Digital Asset Library and Benchmarks Towards High-fidelity Head Avatars.” **NeurIPS 2023 (Joint Second Author)**.

Book Chapter

- **Yuxin Wang**, Linsen Song, Wayne Wu, Chen Qian, Ran He, and Chen Change Loy. “Talking Faces: Audio-to-Video Face Generation.” *Handbook of Digital Face Manipulation and Detection*. **Springer 2022**.

SKILLS and SERVICES

- **Research Areas:** Vision-Language Models, Visual Grounding, GUI Agents, 3D Vision.
- **Programming:** Python, C/C++, Matlab, LaTeX.
- **Languages:** Mandarin Chinese, English.
- **Conference Reviewer:** CVPR, ICCV, ICLR, NeurIPS, AAAI, and ACM MM.
- **Teaching Assistant:** Fundamentals of Artificial Intelligence; Introduction to Object-oriented Programming; Deep Perception, Localization, and Planning for Autonomous Vehicles.